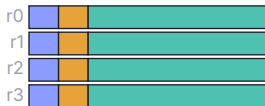


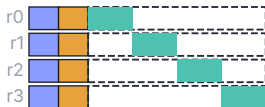
DDP

16 B/param



ZeRO-1

7 B/param



ZeRO-2

5.5 B/param



ZeRO-3

4 B/param



■ bf16 params: 2 bytes/param
■ bf16 grads: 2 bytes/param
■ fp32 optimizer state: 12 bytes/param